Python Programming (CST-005) - Unit 5

Unit 5 - File Handling - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: file handling - reading from and writing to files, plus the usual exceptions around them.

Key ideas: Open a file with **open(name, mode)**: 'r' read (default, error if missing), 'w' write (creates/truncates), 'a' append, 'r+' read+write; add 'b' for binary. The **with open(...) as f:** block closes the file automatically - always prefer it. Read with read(), readline(), readlines() (list of lines) or just loop over f. Write with write()/writelines(). seek(0) moves the file pointer back to the start; tell() says where it is. Wrap file code in try/except to catch **FileNotFoundError** and IOError. CSV and JSON files appear often: the csv and json modules (json.load/json.dump) handle them.

Exam pointers: the standard program: read a text file and count lines/words/characters - practise it with the with-block. Short theory: file modes table, "why prefer with", and read vs readline vs readlines differences.

Made by Anas. Free forever. Watch the linked videos on the site after reading.